

OVAR AB, Portugal

WMO: 085440

Lat: 40.9158N

Lon: 8.6411W

Elev: 72

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.8	27.3	21.3	38.9	30.1	24.3	38.5	22.6	53.9	20.5	54.6	1.9	0	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	84.5	66.4	80.7	65.5	77.1	64.9	69.6	77.6	68.4	75.4	67.2	73.4	8.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	101.6	72.3	66.2	96.9	70.5	64.8	92.2	69.3	33.5	78.8	32.5	75.0	31.6	73.2	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.2	16.9	27.8	94.7	2.4	3.6	26.1	97.3	24.7	99.5	23.4	101.5	21.7	104.2		
(5)			26.9	72.2	2.4	1.8	25.2	73.5	23.7	74.5	22.4	75.6	20.6	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	49.6	50.1	53.3	56.6	60.3	63.8	66.2	66.3	64.9	62.3	54.6	50.6		
	DBStd	7.71	4.75	4.54	4.22	4.61	4.69	4.46	3.76	4.03	4.04	4.95	5.12	5.32		
	HDD50	212	68	51	16	4	0	0	0	0	0	0	13	60		
	HDD65	2717	477	418	363	255	162	72	24	26	51	111	311	447		
	CDD50	3232	56	54	119	203	319	415	502	506	447	381	152	78		
	CDD65	260	0	0	0	4	16	36	61	67	49	27	0	0		
	CDH74	1560	0	0	7	40	146	264	313	356	269	164	1	0		
	CDH80	426	0	0	0	8	33	76	122	107	49	31	0	0		
(14) Wind	WSAvg	6.9	6.0	7.0	7.5	7.6	7.9	7.8	7.9	7.3	6.3	6.2	5.8	5.4		
(15) Precipitation	PrecAvg	49.6	6.9	4.4	3.9	4.3	3.6	1.5	0.7	1.2	3.1	6.7	6.8	6.5		
	PrecMax	69.5	17.6	12.5	22.2	14.8	8.6	4.5	2.7	5.9	8.2	13.1	16.3	18.4		
	PrecMin	26.9	0.6	0.0	0.0	1.3	0.0	0.0	0.1	0.0	0.4	0.9	0.0	0.0		
	PrecStd	10.5	4.4	3.2	4.4	3.1	2.3	1.3	0.6	1.3	2.2	3.0	3.9	3.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	68.0	73.8	78.8	84.4	87.7	90.0	89.4	84.7	83.9	72.5	66.5		
		MCWB	56.5	56.3	57.1	63.1	65.0	68.1	70.5	69.6	65.1	65.6	61.4	57.8		
	2%	DB	60.7	62.8	68.6	73.1	78.3	80.9	82.5	82.4	80.6	78.5	68.4	63.1		
		MCWB	56.5	54.6	56.3	60.0	63.6	65.1	67.7	66.0	64.9	64.7	60.4	56.9		
	5%	DB	59.2	60.1	64.7	68.4	73.1	75.6	75.4	77.2	76.9	74.5	65.8	61.2		
		MCWB	55.9	54.6	55.0	58.8	62.2	64.3	66.4	65.9	64.6	63.6	59.7	56.6		
	10%	DB	57.6	58.7	61.2	64.8	68.4	71.5	72.2	73.6	73.4	71.3	63.2	59.3		
		MCWB	54.7	54.3	54.9	58.1	60.4	63.6	65.5	65.3	64.9	63.6	58.9	55.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	59.2	60.1	63.6	67.5	69.9	72.7	72.1	69.6	69.3	64.6	61.0		
		MCDB	62.1	61.3	65.9	74.6	79.3	81.6	86.2	81.4	76.0	76.3	68.0	62.9		
	2%	WB	58.4	57.8	58.5	61.5	65.4	67.3	69.1	69.5	68.1	67.5	62.8	59.2		
		MCDB	59.7	60.2	63.2	69.6	74.4	76.8	77.7	76.2	74.8	72.7	66.4	61.7		
	5%	WB	57.1	56.3	57.4	60.1	63.4	65.7	67.5	68.2	67.0	66.3	61.4	58.0		
		MCDB	58.8	59.3	61.6	66.7	70.5	72.6	73.9	73.8	73.1	71.0	64.6	60.7		
	10%	WB	55.8	54.6	56.2	58.9	61.7	64.4	66.4	66.9	65.6	65.0	59.8	56.4		
		MCDB	57.7	58.0	60.3	64.4	67.0	69.8	71.4	72.0	71.2	69.8	62.7	59.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	16.1	15.6	15.2	15.0	14.5	13.8	15.8	16.3	17.0	16.5	16.9		
		MCDBR	15.2	18.6	23.8	22.5	23.6	23.9	21.8	22.7	23.2	22.9	18.2	16.5		
		MCWBR	11.6	12.4	13.7	12.1	11.5	10.5	10.2	10.1	10.4	12.0	11.7	11.9		
	5% WB	MCDBR	9.9	13.7	16.2	18.4	19.1	18.7	16.3	17.2	17.4	16.5	13.4	12.2		
		MCWBR	8.8	10.6	11.1	10.9	10.3	9.4	8.6	9.1	9.4	9.8	10.0	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.345	0.378	0.378	0.385	0.390	0.380	0.393	0.381	0.371	0.343	0.331		
	taud		2.503	2.462	2.375	2.387	2.393	2.397	2.415	2.376	2.431	2.473	2.506	2.509		
	Ebn at Noon		260	275	276	284	283	281	282	275	272	261	255	252		
	Edh at Noon		24	29	35	37	37	37	36	37	33	28	24	22		
(44) All-Sky Solar Radiation	RadAvg		603	928	1270	1669	2035	2216	2264	2047	1597	1036	683	547		
	RadStd		100	128	180	191	180	144	110	100	115	116	115	81		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83

Nomenclature: See separate page